Before Learning from Cellular Traces: Auditing Throughput Semantics and Measurement Timing in a Public 5G/LTE Dataset

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Abstract

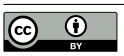
Public cellular traces are widely used for machine-learning studies, yet their measurement semantics and timing require careful validation. We audit downlink throughput measurements in a public 5G/LTE dataset by reconstructing reported throughput from received-byte counts using an empirically inferred nominal 0.5-s accounting duration. Across two T-Mobile collection days, the reconstruction shows strong agreement with reported throughput (Pearson correlations of 0.999752 and 0.999712), with a median absolute residual of 0.000368 Mbps on both days. Residuals exceeding 5 Mbps occur in only 0.653% and 0.574% of observations, respectively. Threshold-sensitivity and traffic-rate analyses show that the magnitude and localization of exceptional residuals depend on the chosen absolute-error threshold. Although the central accounting relationship replicates, the timing and interval-level patterns of departures differ across days. These findings demonstrate the importance of auditing measurement semantics before interpreting cellular traces or using them for predictive modeling.

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1 Introduction

Public cellular-network traces are increasingly used to characterize mobile-network performance and develop machine-learning models for throughput prediction [2, 3]. Recent measurement studies have also released large-scale multi-operator 5G/LTE traces collected under mobility [1]. However, reported throughput and related measurements may reflect undocumented aggregation or reporting behavior. Without validating these semantics, measurement artifacts may be mistaken for network dynamics or incorporated into predictive models.

We audit downlink throughput measurements in a public 5G/LTE dataset by reconstructing reported throughput from separately logged received-byte counts. We empirically infer a nominal 0.5-s accounting duration and examine reconstruction residuals, reporting intervals, and sensitivity to absolute-error thresholds and traffic rates. We further replicate the analysis on a second collection day to assess the stability of the observed relationship.



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Our results show that the byte–throughput relationship is highly consistent across both days, while a small fraction of large residuals exhibits varying temporal and interval-level patterns. These findings motivate measurement-semantic validation before interpreting cellular traces or using them for predictive modeling.

2 Dataset and Audit Method

We analyze T-Mobile downlink (DL) measurements from Drive 2, Days 1 and 2 of the public PAM 2025 multi-carrier dataset [1]. Using the author-provided DL test intervals and reproduced timestamp encoding, we identify 12,706 target-bearing observations across 55 DL intervals on Day 1 and 11,159 observations across 47 intervals on Day 2.

We assess internal consistency by reconstructing reported DL throughput from the separately logged received-byte field. The dominant byte–throughput relationship empirically supports a nominal 0.5-s accounting duration:

$$\hat{T} = \frac{8B}{0.5 \times 10^6}, \tag{1}$$

where B is the received-byte quantity and $\hat{T}$ is reconstructed throughput in Mbps. We compare $\hat{T}$ with reported throughput T using the absolute residual $e = |\hat{T} - T|$.

The nominal 0.5-s duration is supported by two complementary diagnostics: observed report spacing and the accounting duration implied by byte-throughput ratios. We examine residuals above 5 Mbps as a descriptive threshold, not a physical validity boundary. To assess sensitivity, we vary the threshold and examine residual behavior across traffic-rate groups. We also inspect timing and interval-level patterns of exceptional observations and compare their characteristics across the two collection days.

3 Results

Reconstruction and replication. The nominal 0.5-s reconstruction closely matches reported DL throughput on both collection days (Table 1). On Day 1, Pearson's r is 0.999752 across 12,706 observations; on Day 2, r is 0.999712 across 11,159 observations. Both days have a median absolute residual of 0.000368 Mbps. On Day 1, 98.18% of consecutive target reports have 0.5-s spacing, and all 55 DL intervals have median implied accounting durations between 0.49 and 0.51 s. These results support a reproducible nominal accounting relationship rather than an assumed logger implementation.

Threshold and rate sensitivity. On Day 1, 83 observations (0.653%) exceed a 5-Mbps absolute residual, compared with 64 (0.574%) on Day 2. Threshold sensitivity is substantial: the Day-1 count decreases from 899 at 1 Mbps to 31 at 50 Mbps. The share of flagged observations in Day-1 interval 6 rises from 6.2% to 100% across these thresholds. Upper-tail absolute residuals also increase

Table 1: DL throughput reconstruction across two collection days.

Metric	Day 1	Day 2
DL observations	12,706	11,159
DL intervals	55	47
Pearson r	0.999752	0.999712
Median absolute residual (Mbps)	0.000368	0.000368
Residuals > 5 Mbps	83 (0.653%)	64 (0.574%)
Affected intervals	16	17
Top-interval share (> 5 Mbps)	51.8%	29.7%

with traffic rate, demonstrating that a fixed absolute threshold is not rate-neutral.

Exceptional observations. The 83 Day-1 residuals above 5 Mbps occur across 16 DL intervals, with 43 in interval 6. Detailed diagnostics reveal heterogeneous timing behavior: 38 of the 40 departures outside interval 6 have nominal preceding report spacing. The remaining departures cannot be assigned to a unique mechanism using the available measurements. Day 2 reproduces the small exceptional tail, but its interval-level and timing patterns differ from Day 1.

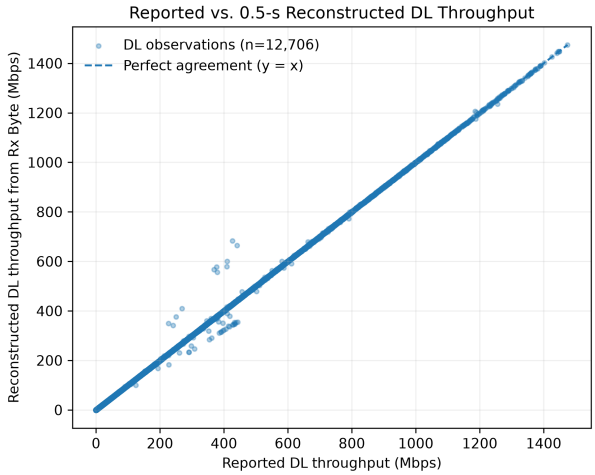


Figure 1: Reported versus reconstructed DL throughput across 12,706 audited observations. The dashed line indicates perfect agreement ($y = x$).

4 Discussion and Conclusion

Across two collection days, the received-byte and reported-throughput fields exhibit a reproducible nominal 0.5-s accounting relationship. However, the small residual tail varies in its timing and interval-level distribution. Although some departures coincide with irregular reporting intervals, the available evidence does not establish a unique causal mechanism.

Our results demonstrate that absolute-error thresholds are sensitive to traffic rate and should not be interpreted as physical validity boundaries. The audit provides an internal consistency check rather than independent throughput ground truth.

The predictive impact of these departures remains unmeasured. Our existing ML benchmark uses a nonoverlapping August 2022 trace, preventing a valid row-level ablation on the audited May 2023 observations. Future work will examine this impact using leakage-safe validation on the audited cohort.

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